Two-Equation Turbulence Models

Problem

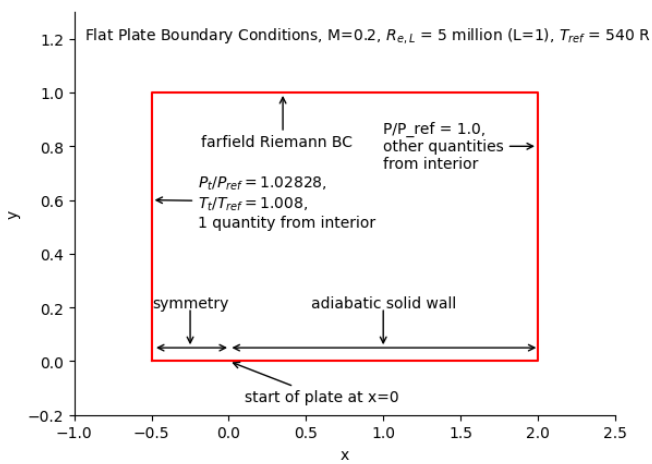
The objective of this study is to model the turbulent boundary layer over a 2D flat plate using a Computational Fluid Dynamics (CFD) solver ANSYS Fluent employing the two equation model. In this exercise, the two-equation turbulence models, standard $k - \varepsilon$ and $k - \omega$, will be used. This simulation will validate the CFD results against theoretical predictions by comparing:

- Wall skin friction coefficient (C_f) at different momentum-thickness Reynolds numbers ($4000 < R_{e,\theta} < 13000$).
- The law of the wall at $R_{e,\theta} = 10000$ to verify the near-wall velocity distribution.
- In addition, one should also calculate results using two different grid resolutions and show the distributions of k, ω, ε and μ.

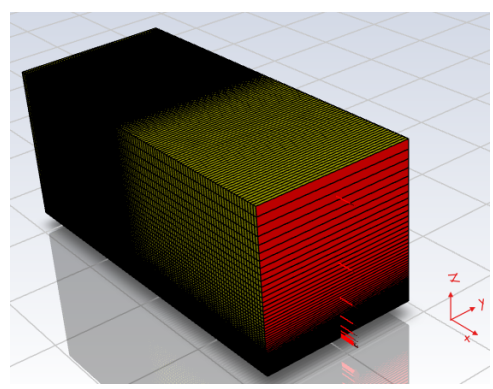
The aim is to evaluate the accuracy of the two-equation models and their ability to capture essential flow characteristics in a turbulent boundary layer.

Solution

The study of boundary layer flow over a flat plate is fundamental in fluid mechanics, particularly in understanding transition from laminar to turbulent flow.



(a) 2D flat surface with boundary conditions.



(b) Mesh representation in Fluent.

This problem is relevant in aerodynamics, heat transfer, and engineering applications where surface friction and turbulence effects play a crucial role in system performance. Computational Fluid Dynamics (CFD) provides a powerful tool to analyze such flows by solving the Navier-Stokes equations with turbulence models that approximate the behavior of small-scale eddies. Boundary conditions along with 2D representation of the plate is presented on the Figure 1a.

Setup

At the inlet, a uniform free-stream velocity U_∞ is prescribed to simulate the flow approaching the flat plate. Turbulence intensity is set based on the Reynolds number, ensuring accurate boundary layer development. A pressure

outlet boundary condition is applied at the downstream edge of the computational domain, with the pressure set to zero gauge pressure. This allows for a smooth exit of the fluid, reducing artificial reflections. The flat plate surface is modeled using a no-slip boundary condition, where both the streamwise and normal velocities are set to zero. This enforces the correct shear stress distribution necessary for boundary layer formation. A far-field boundary condition is applied to the upper edge of the domain. This allows the flow to maintain free-stream conditions, with the velocity set to U_∞ and pressure set to ambient values.

A CFD solver is used to simulate the flow with the following settings:

- Turbulence Model: standard $k-\omega$, standard $k-\varepsilon$
- Grid: Structured mesh with refinement near the wall, coarser mesh 273×193 and finer mesh 545×385
- Solver: Pressure-based, steady-state
- Post-processing: Skin friction coefficient and k, ω, μ, ε distributions

Definition of the quantities are given in the set of equations below.

$$Re_\theta = \frac{\rho_\infty U_\infty \theta}{\mu_\infty} \tag{1}$$

$$\theta = \int_0^\infty \frac{\rho}{\rho_\infty}\frac{u}{U_\infty}\left(1-\frac{u}{U_\infty}\right)dy \tag{2}$$

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_\infty U_\infty^2} \tag{3}$$

$$\tau_w = \mu_w \left(\frac{\partial u}{\partial y}\right)_w \tag{4}$$

$$u^+ = \frac{u}{v^*} = \frac{u}{\sqrt{\tau_w/\rho_w}} \tag{5}$$

$$y^+ = \frac{y v^*}{\nu_w} = \frac{y\sqrt{\tau_w/\rho_w}}{\nu_w} \tag{6}$$

Two-equation Turbulence Models

Standard $k-\varepsilon$

The turbulence kinetic energy, k, and its rate of dissipation, ε, are obtained from the following transport equations: In these equations, G_k represents the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. $C_{1\varepsilon}$, $C_{2\varepsilon}$, and $C_{3\varepsilon}$ are constants. σ_k and σ_ε are the turbulent Prandtl numbers for k and ε, respectively. and are user-defined source terms.

$C_{1\varepsilon}$=1.44, $C_{2\varepsilon}$=1.92, and C_μ=0.09, σ_k=1.0 and σ_ε=1.3 default values have been determined from experiments for fundamental turbulent flows including frequently encountered shear flows like boundary layers, mixing layers and jets as well as for decaying isotropic grid turbulence. They have been found to work fairly well for a wide range of wall-bounded and free shear flows.

$$\frac{\partial}{\partial t}(\rho k)+\frac{\partial}{\partial x_i}(\rho k u_i)=\frac{\partial}{\partial x_j}\left[\left(\mu+\frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right]+G_k+G_b-\rho\varepsilon-Y_M+S_k$$

$$\frac{\partial}{\partial t}(\rho \varepsilon)+\frac{\partial}{\partial x_i}(\rho \varepsilon u_i)=\frac{\partial}{\partial x_j}\left[\left(\mu+\frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right]+C_{1\varepsilon}\frac{\varepsilon}{k}\left(G_k+C_{3\varepsilon}G_b\right)-C_{2\varepsilon}\rho\frac{\varepsilon^2}{k}+S_\varepsilon$$

The turbulent viscosity, ν_t, is computed by combining as follows:

$$\nu_t = \rho C_\mu \frac{k^2}{\varepsilon} \tag{7}$$

Standard $k-\omega$

The turbulence kinetic energy, k, and the specific dissipation rate, ε, are obtained from the following transport equations:

$$\frac{\partial}{\partial t}(\rho k)+\frac{\partial}{\partial x_i}(\rho k u_i)=\frac{\partial}{\partial x_j}\left(\Gamma_k\frac{\partial k}{\partial x_j}\right)+G_k-Y_k+S_k+G_b$$

$$\frac{\partial}{\partial t}(\rho \omega)+\frac{\partial}{\partial x_i}(\rho \omega u_i)=\frac{\partial}{\partial x_j}\left(\Gamma_\omega\frac{\partial \omega}{\partial x_j}\right)+G_\omega-Y_\omega+S_\omega+G_{\omega b}$$

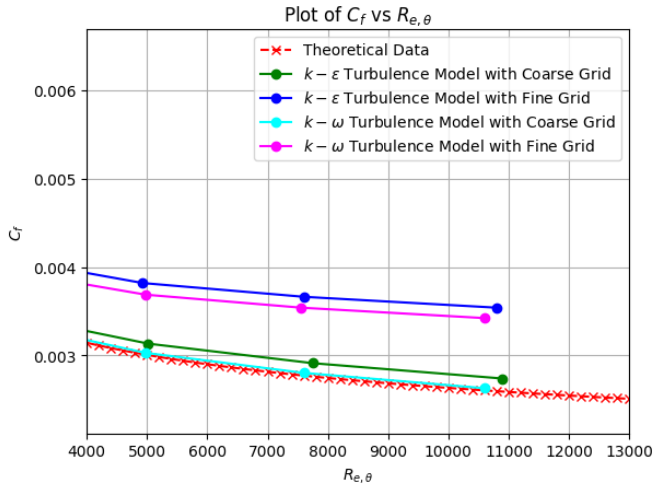
In these equations, G_k represents the generation of turbulence kinetic energy due to the mean velocity gradients. G_ω represents the generation of ω. Γ_k and Γ_ω represent effective diffusivity of k and ω, respectively. Y_k and Y_ω represent the dissipation of k and ω due to the turbulence. S_k and S_ω are user-defined source terms. G_b and $G_{\omega b}$ account for buoyancy.

The turbulent viscosity, ν_t, is computed by combining as follows:

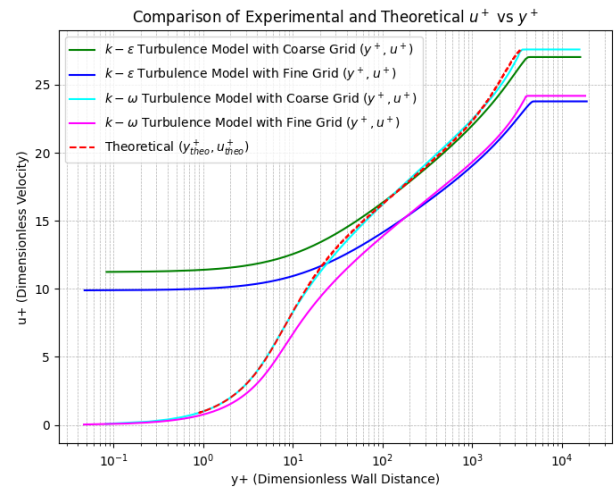
$$\nu_t = \alpha^* \frac{\rho k}{\omega}, \qquad \alpha^* = 1.0. \tag{8}$$

Results & Discussion

Distributions of the wall skin friction coefficient C_f over different Reynolds numbers against theoretical values for this case are presented on Fig. 2a. The Law of the Wall for different turbulence models against theoretical values is presented on the Fig. 2b.



(a) The wall skin friction coefficient over the Reylonds number for the range $4000 < R_e < 13000$ for different turbulence models using two different meshes.



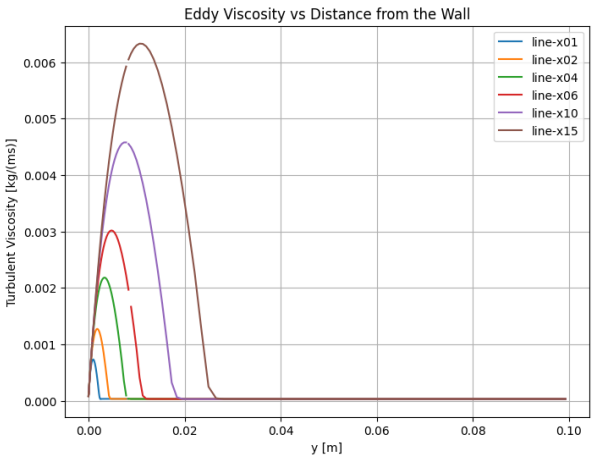
(b) The law of the wall distributions for different turbulence models applied on two different meshes (R_e=10000).

From Fig. 2a can be seen that Standard $k-\omega$ and $k-\varepsilon$ two-equation turbulence models on coarser grid (273×139) reproduce more faithful values of C_f rather than on the finer grid (545×385) comparing to the theoretical values. Between $k-\omega$ and $k-\varepsilon$ on the coarse grid, $k-\omega$ gives the closest fit to the theoretical predictions. This can be explained by the models grid sensitivity and near-wall treatment: the $k-\omega$ mode is known for its better near-wall resolution becase ω governs the turbulent length scale close to the wall. On the other hand the $k-\varepsilon$ model is more accurate in the fully turbulent region but requires wall functions for near-wall treatment, which can introduce errors on finer grids if the first grid point is not placed correctly in the boundary layer. One other point is, in the coarse grid, the gradients may not be perfectly captured, but wall functions compensate for this, making the results closer to the theoretical predictions. In the fine grid, if the first cell height is not sufficiently small, the turbulence model might not handle the boundary layer correctly, leading to deviations.

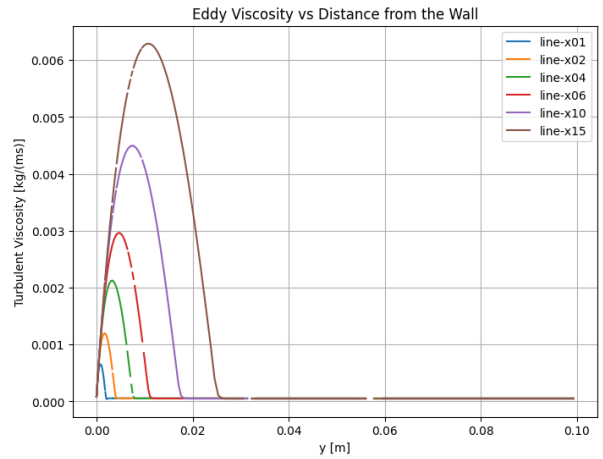
The explanations from above are, actually, confirmed on Fig. 2b. The law of the wall is the best reproduced with $k-\omega$ model on the coarser grid (line in cyan). $k-\varepsilon$ model on both grids give very bad results in the viscous sublayer, but $k-\varepsilon$ model on coarse grid captures logarithmic layer behavior very well. It is because the $k-\varepsilon$ model is originally designed for high-Reynolds-number turbulence and does not perform well for small distances from the wall. However, on the fine grid, the incorrect near-wall resolution affects the entire profile, leading to deviations in both the viscous and logarithmic layers for both models. A finer grid should improve accuracy, but only if the turbulence model is correctly formulated for low y^+. If the grid is too fine for wall functions, but not fine enough for direct resolution of the viscous layer ($y^+ \approx 1$), the model fails to provide correct near-wall predictions.

Distributions of k, ϵ, ω and ν_t

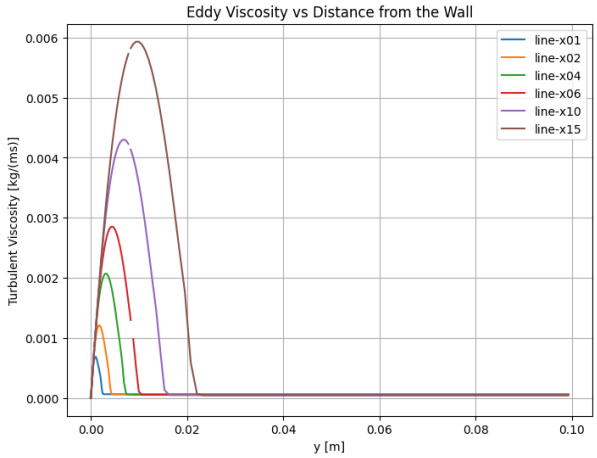
As part of the analysis of this case, the distributions of eddy viscosity for both models on both grids are calculated and presented on Fig. 3. The eddy viscosity ν_t represents the additional viscosity caused by turbulence, which enhances the momentum transfer. In the viscous sublayer, the flow is laminar-like, dominated by molecular viscosity, and turbulence is very weak. Since turbulence production is minimal in this region, $\nu_t \approx 0$ near the wall.
In the buffer layer and logarithmic layer, turbulence production increases due to shear effects. The peak in turbulence kinetic energy and turbulence dissipation occurs in this region. The maximum eddy viscosity is found at



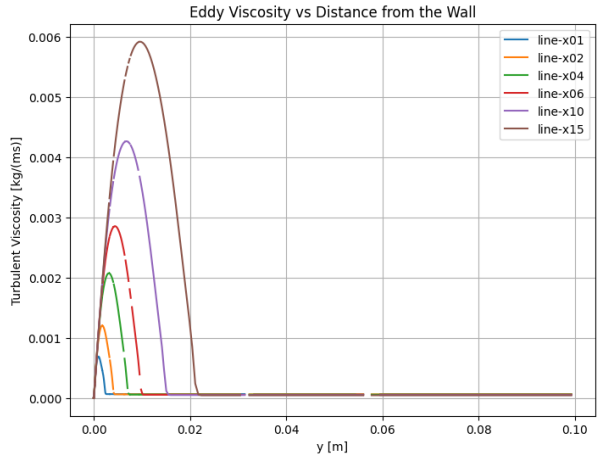
(a) Eddy Viscosity, ν_t, distribution for $k-\varepsilon$ turbulence model on coarse mesh.



(b) Eddy Viscosity, ν_t, distribution for $k-\varepsilon$ turbulence model on fine mesh.



(c) Eddy Viscosity, ν_t, distribution for $k-\omega$ turbulence model on coarse mesh.



(d) Eddy Viscosity, ν_t, distribution for $k-\omega$ turbulence model on fine mesh.

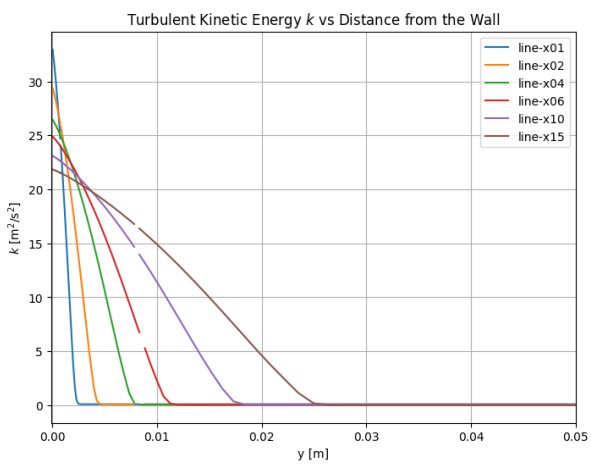
Figure 3: Eddy viscosity, μ_t, distributions for $k-\omega$ and $k-\varepsilon$ turbulence models for both grids.

the location where turbulence production is highest. As you move further away from the wall into the free-stream region, velocity gradients become small. Turbulence generation diminishes, and eddy viscosity gradually decreases to zero. In the fully developed outer region, turbulence is mainly convected away, and molecular viscosity dominates again.

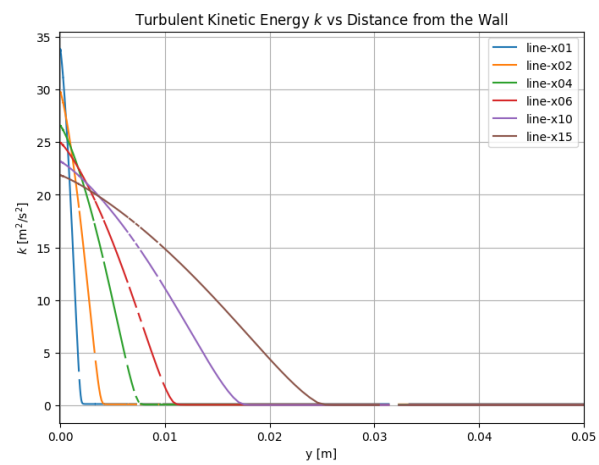
The maximum eddy viscosity is generally higher in the $k-\varepsilon$ model due to its dependence on the turbulent dissipation rate (which can be relatively small in high turbulence regions), allowing for larger values. On the other hand, the $k-\omega$ model typically predicts a more tightly controlled eddy viscosity, especially near walls, due to its dependence on the specific dissipation rate, which tends to remain higher in such regions. Finer grid doesn't affect eddy viscosity calculation.

Turbulence kinetic energy distributions are presented on Fig. 4. In turbulent flows near solid boundaries, such as

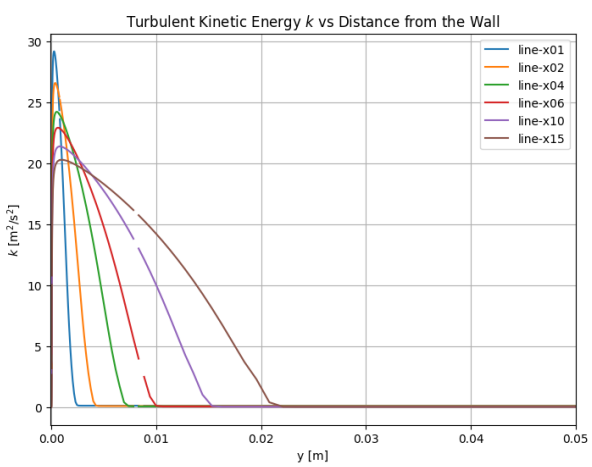
the walls of a pipe or the surface of a flat plate, the turbulence is strongly influenced by the interaction between the fluid and the wall. The boundary layer is a region of the flow where the velocity gradient is high due to the **no-slip condition** at the wall. Close to the wall, turbulent eddies are generated and break down at small scales due to friction. The turbulent kinetic energy is created in this region by the large vortical structures and dissipates into smaller scales as it moves away from the wall. Turbulent kinetic energy is highest close to the wall because of the intense shear stress and turbulence generation in that region. As you move away from the wall, the velocity gradients decrease, reducing the turbulence intensity. However, the energy from turbulent eddies takes time to dissipate, which leads to the slower decrease of turbulent kinetic energy as you move farther from the wall.



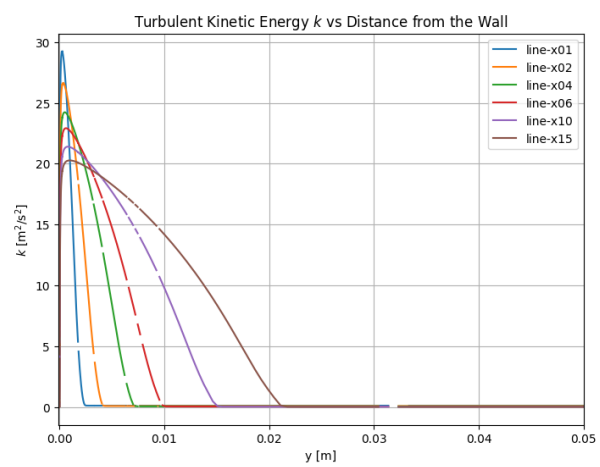
(a) Turbulence kinetic energy, k, distribution for $k-\varepsilon$ turbulence model on coarse mesh.



(b) Turbulence kinetic energy, k, distribution for $k-\varepsilon$ turbulence model on fine mesh.



(c) Turbulence kinetic energy, k, distribution for $k-\omega$ turbulence model on coarse mesh.



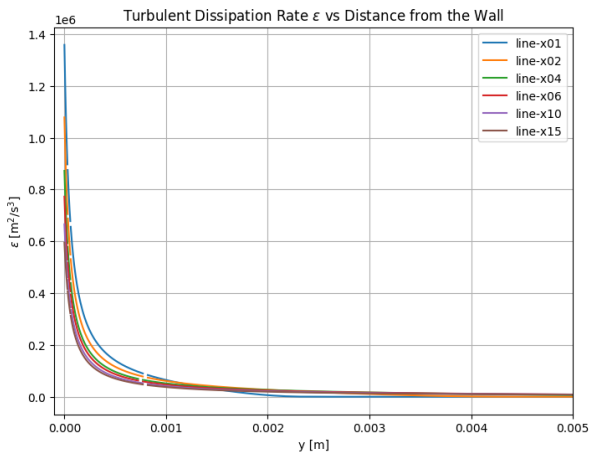
(d) Turbulence kinetic energy, k, distribution for $k-\omega$ turbulence model on fine mesh.

Figure 4: Turbulence kinetic energy, k, distributions for $k-\omega$ and $k-\varepsilon$ turbulence models for both grids.

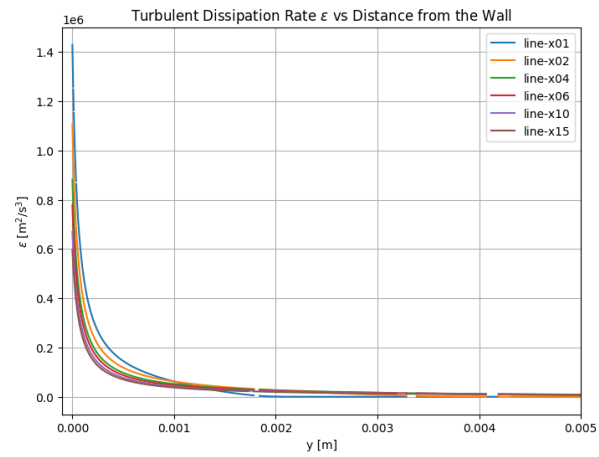
Since the $k-\omega$ model is more sensitive to near-wall effects, k starts from zero at the wall, reflecting the absence of turbulence there due to the no-slip boundary condition. The turbulence kinetic energy increases as you move away from the wall, with the growth controlled by the ω term. In $k-\varepsilon$ model, the turbulent kinetic energy starts from a positive value at the wall because the model assumes that turbulence is present in the boundary layer (even close

to the wall) and is generated by the shear stress from the no-slip condition. Finer mesh produces little bit higher maximum turbulent kinetic energy, but the results for both meshes are consistent.

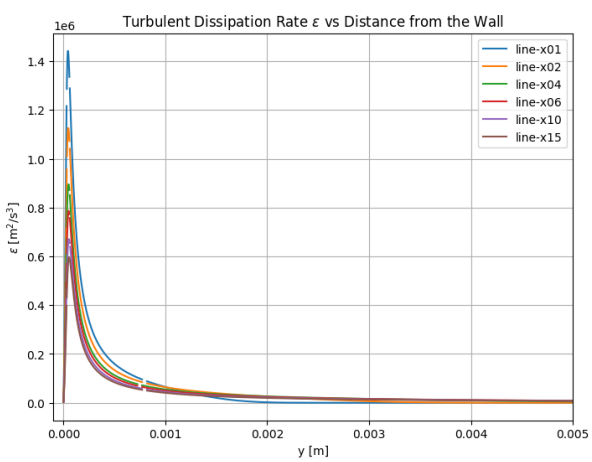
The turbulence dissipation rate distributions, ε, are presented on the Fig. 5. In both models, the dissipation rate ε peaks in the near-wall region where turbulence production is highest and then gradually decreases away from the wall. Farther from the wall, turbulence is more homogeneous, and large eddies dominate over smaller dissipative eddies. Since dissipation mainly occurs at the smallest turbulence scales, and these scales are reduced farther from the wall, ε decreases until it reaches zero. In $k-\varepsilon$ model, the wall function approximation assumes that turbulence is well-developed in the near-wall region, even though the mesh does not resolve the smallest eddies, so ε remains finite because k is nonzero at the wall. In $k-\omega$ model, the dissipation rate ε is zero at the wall because both k and ω are defined in such a way that the dissipation physically vanishes where turbulence is absent. The finer mesh didn't affect the calculation, and the results for different grids for both models are consistent.



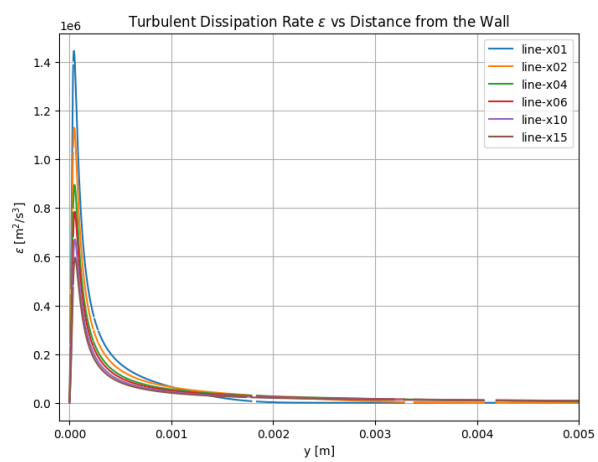
(a) Turbulence dissipation rate, ε, distribution for $k-\varepsilon$ turbulence model on coarse mesh.



(b) Turbulence dissipation rate, ε, distribution for $k-\varepsilon$ turbulence model on fine mesh.



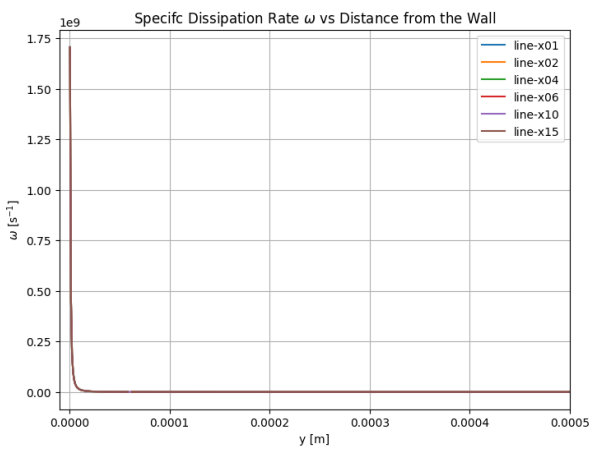
(c) Turbulence dissipation rate, ε, distribution for $k-\omega$ turbulence model on coarse mesh.



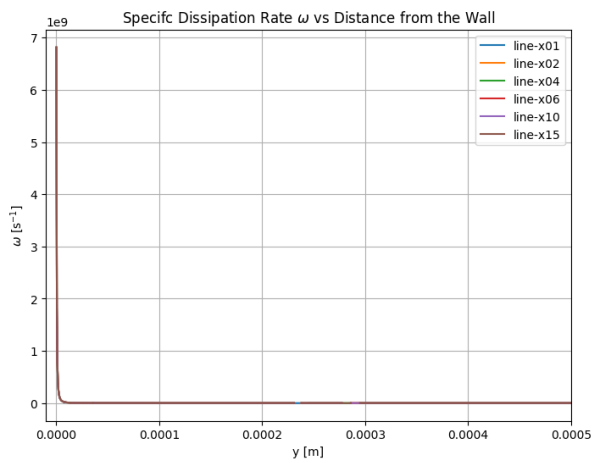
(d) Turbulence dissipation rate, ε, distribution for $k-\omega$ turbulence model on fine mesh.

Figure 5: Turbulence dissipation rate, ε, distributions for $k-\omega$ and $k-\varepsilon$ turbulence models for both grids.

The distributions of specific dissipation rate are presented on Fig. 6. As it can be seen, the specific dissipation rate starts from a high value and rapidly decreases, and it is only present in $k-\omega$ model. Near the wall, ω needs to be very large because: k is very small (since turbulence production is limited by viscosity), ε is even smaller (since there is little turbulence dissipation in the viscous sublayer). Since$\omega = \varepsilon/k$, if both ε and k are small but ε is relatively smaller, ω becomes very large. After moving slightly away from the wall, k increases faster than ε, so ω decreases very quickly. It can be observed that finer mesh produces higher initial value of the ω. The first cell height (y^+) in the mesh controls how well the boundary layer is resolved. For fine mesh the first grid point is very close to the wall, meaning the solver captures more of the steep gradients in turbulence quantities. Since ω is inversely proportional to the square of the near-wall distance, a smaller y directly leads to a higher initial value of ω. If the mesh is too coarse, the ω peak is underpredicted, leading to inaccurate turbulence modeling in the near-wall region. If the mesh is too fine, it may lead to numerical stiffness in the solver due to very high ω values.



(a) Specific dissipation rate, ω, distribution for $k-\omega$ turbulence model on coarse mesh.



(b) Specific dissipation rate, ω, distribution for $k-\omega$ turbulence model on fine mesh.

Figure 6: Specific dissipation rate, ω, distributions for $k-\omega$ and $k-\varepsilon$ turbulence models for both grids.

General remark: Through all the plots, the values plotted are taken at the certain x-position (x=0.1, 0.2, 0.4, 0.6, 1.0, 1.5). It will be evident that values associated with higher x value, will be higher. It can be interpreted by understanding that turbulence needs some distance to fully develop. At lower x values (closer to the inlet), turbulence is still growing, and ε and ω are lower. As x increases, the boundary layer thickens, shear increases, and turbulence production intensifies, leading to higher dissipation rates.

Conclusion

This study highlights the importance of grid resolution and turbulence model selection in CFD simulations. While finer grids theoretically improve accuracy, improper near-wall treatment can lead to unexpected deviations. The $k-\omega$ model is preferable for near-wall accuracy, while the $k-\varepsilon$ model performs better in the fully turbulent region.